WOMEN'S DIABETES RISK PREDICTION SYSTEM

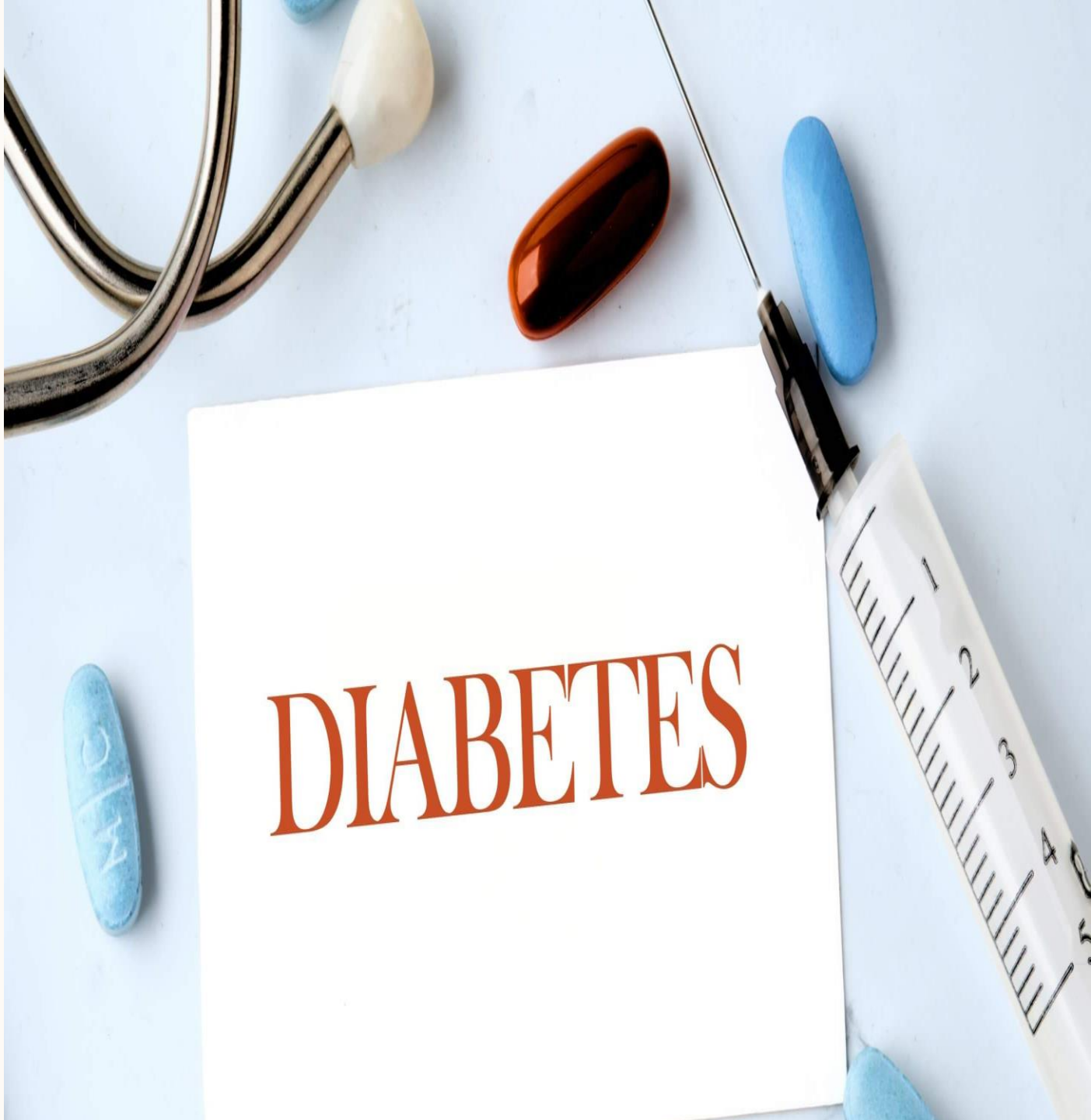
AGENDA

Problem Statement

Proposed solution

Methodology

System design





PROBLEM STATEMENT

DIABETES IS A MAJOR PUBLIC HEALTH CONCERN GLOBALLY, ESPECIALLY IN DEVELOPING REGIONS LIKE SOUTH AFRICA. WOMEN FACE UNIQUE RISKS RELATED TO DIABETES, INFLUENCED BY BIOLOGICAL, LIFESTYLE, AND SOCIO-ECONOMIC FACTORS. MILLIONS OF INDIVIDUALS WORLDWIDE SUFFER FROM DIABETES THESE DAYS, WITH WOMEN HEAVILY AFFECTED BY THE CONDITION [1].

PROPOSED SOLUTION

Develop a system that:

Predicts diabetes risk specifically for women using machine learning.
Uses easily obtainable health indicators like glucose, BMI, age, blood pressure etc.
Employs Random Forest Classifier for robust predictions
Provides feature importance analysis for medical insights



METHODOLOGY

Data Collection:

- Pima Indian Diabetes Dataset (PIDD)
- Female participants only

Feature selection:

- 8 key health indicators
- Glucose level (primary predictor)
- BMI, Age, Diabetes Pedigree Function
- Blood pressure, Pregnancies, Skin thickness, Insulin

Machine learning model:

- Random Forest Classifier
- 300 decision trees (n_trees=300)
- Maximum depth of 15 (max_depth=15)
- Stratified splitting for balanced training

DIABETES PEDIGREE FUNCTION

- A measure of genetic influence on diabetes risk
- Summarizes family history of diabetes into a single number
- Higher score → stronger family history → higher risk

Formula (simplified):

- DPF = (Sum of relatives with diabetes × weight)

0 = non-diabetic, 1 = diabetic

Weights based on closeness:

- - Parents/Siblings → 0.5 each
- - Grandparents/Aunts/Uncles/Cousin → 0.25 each

Example Calculation

- Person with:

diabetic parent → 0.5
non-diabetic siblings → 0.5
diabetic grandparent → 0.25

PDF = (1)(0,5) + (0)(0,5) + (1)(0,25) = 0,75 (Moderate)

DPF SCORE INTERPRETATION

- - Low Risk: DPF < 0.5 → Weak family history
- - Moderate Risk: 0.5 ≤ DPF < 1.0 → Some family influence
- - High Risk: 1.0 ≤ DPF < 2.0 → Strong family influence
- - Very High Risk: DPF ≥ 2.0 → Very strong genetic predisposition

IMPLEMENTATION DETAILS

Development Environment:

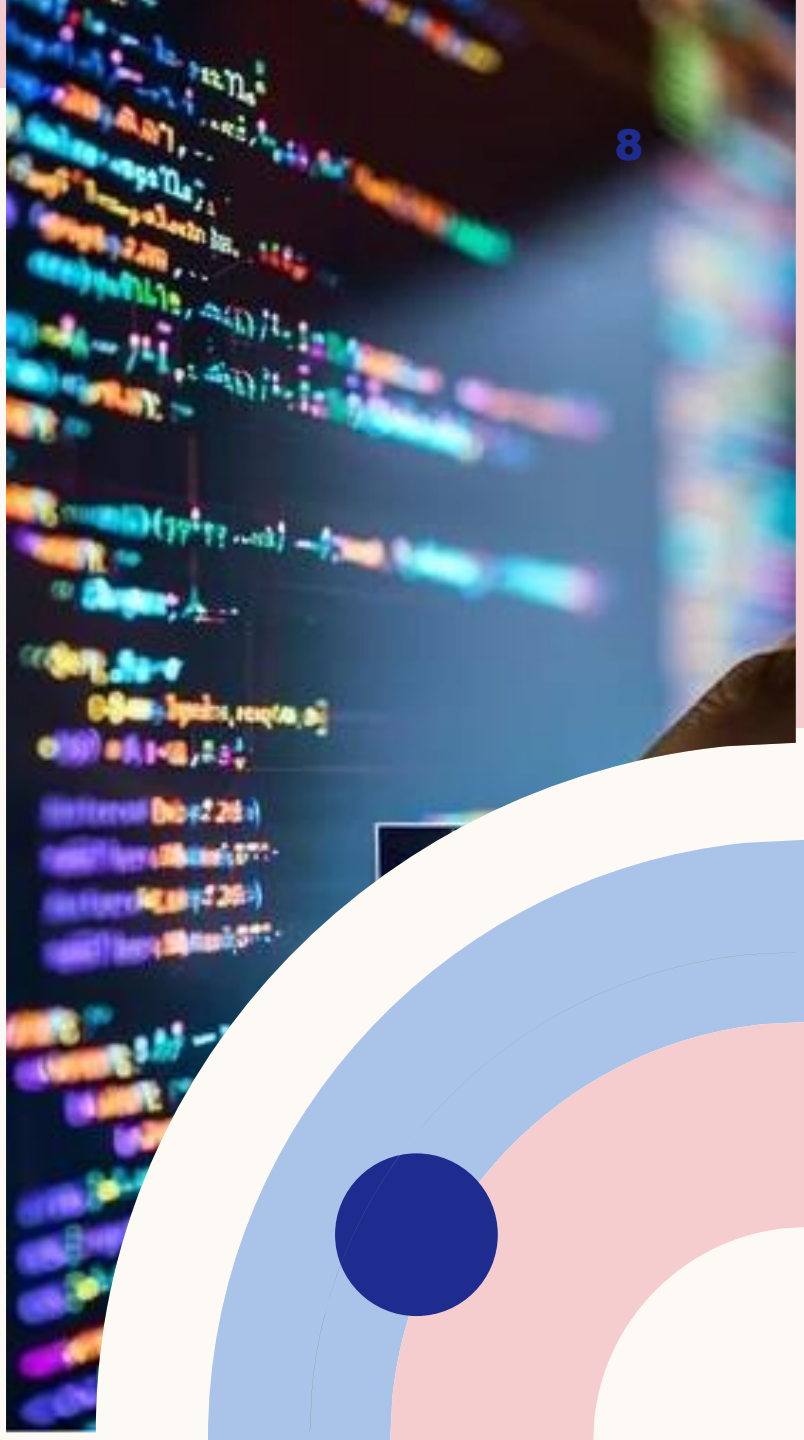
- Python-based implementation
- Numpy and Pandas for model training

Model Configuration:

- Trees: 300 decision trees in ensemble
- Depth: Maximum depth of 15 levels
- Sampling: Bootstrap sampling with replacement

Pipeline Process:

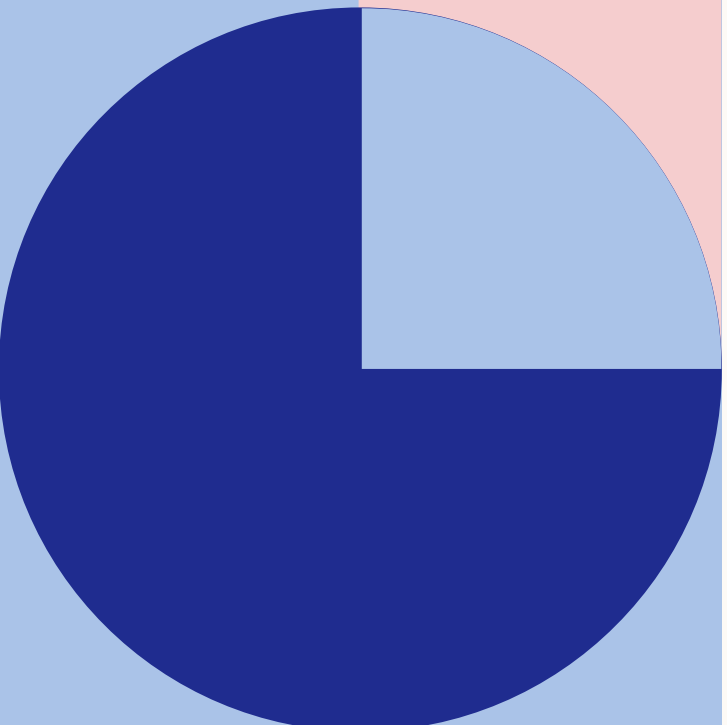
- Data loading and preprocessing
- Stratified train-test split
- SMOTE
- Model training with feature importance calculation
- Performance evaluation and validation



SYSTEM DESIGN

Model Architecture:

- Data Splitting: 80% training, 20% testing with stratification
- Training Pipeline: Bootstrap sampling with replacement
- Feature Importance: Aggregated across all decision trees
- Prediction: Majority voting from ensemble trees



RESULTS

Model Performance Metrics:

Accuracy: 78,29% - Correctly predicts diabetes risk
Precision: 67,24% - When predicting diabetes, 78% accuracy
Recall: 73,58% - Identifies 73,58% of actual diabetes cases
F1 Score: 70,27% - Balanced precision-recall measure

Feature Importance Analysis:

Glucose (43.71%) - Most critical predictor
BMI (19.50%) - Second most important factor
Age (12.84%) - Third significant predictor
Diabetes Pedigree (8.34%) - Genetic risk component
Other factors: Blood pressure, pregnancies, skin thickness, insulin



References:

. Ahmed, Afshan, et al. "Machine Learning Algorithm-Based Prediction of Diabetes among Female Population Using PIMA Dataset." Healthcare, vol. 13, no. 1, 29 Dec. 2024, pp. 37– 37, www.mdpi.com/2227-9032/13/1/37, https://doi.org/10.3390/healthcare13010037.

DEMO !!!!!!!!!!!

DEMO

THANK YOU

Cebo Sithole
218035887
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